

4.1 -> Constraints & Considerations

↳ Considerations & constraints that influence the rate determination. Rate can be broken into three components:

- Regulatory constraints
- Operations - - -
- Accounting considerations

↳ Factors that affect a policyholder's likelihood to renew are:

- Prices of competing products
- Risk level of the product
- Rate changes
- Characteristics of the insured / susceptibility to the risk
- Customer satisfaction & brand loyalty

4.2 -> Final Pricing rates

→ Expense fees & other additions

→ Final premium = $\frac{\text{Base premium} \times \text{relativity}}{\text{variable premium}} + \text{Addition premium}$

→ Expense fee →
$$\bar{F}_p = \frac{\bar{F} + \bar{E}_p + F_p}{1 - v - \bar{q}_p} = \frac{\text{Final expense fee expense}}{\text{VPLR}}$$

$$\downarrow$$

$$= \frac{\bar{F} + \bar{E}_p}{1 - v - \bar{q}_p} + \frac{F_p}{1 - v - \bar{q}_p}$$

variable premium
expense fee (adjusted for VPLR - death premium)

→ Deriving base rates

→ Act. Premium = $\text{Base rate} \times \text{Any rating factor} \rightarrow \text{Addition fee}$

↳ Final step is to derive this

↳ Priority to derive

→ Let the average premium, base rate, weighted average rating factor, & expense fee w/ $\bar{F}, \bar{R}, \bar{q}_p, \bar{F}_p$, respectively, & subscript p to indicate "proposed" & a to indicate "actual"

→ resulting above: $\bar{F}_p = \bar{F} \times \bar{R} + \bar{F}_p$

$\bar{R} = \bar{R} \times \bar{R} + \bar{R}$

* $\bar{R}_p = \frac{\bar{F}_p - \bar{F}}{\bar{F} - \bar{F}_p} = \bar{R} = \frac{\bar{R}}{\bar{R}}$

↳ assuming not changing relativities

⇒ 1

→ Deriving base rates: alternative calculations → three methods

→ Approximated average rate determination

→ The method is called approximated average rate determination because it allows us to approximate the proposed average rating factor (aka proposed rate differential) by using the actual average rating factor. We will approximate the average rating factor as the product of the weighted average rating factors (rate differentials) for each individual rating unit.

→ e.g. Avg rating factor = Avg class factor * Avg Territory factor

→ Weight relativities by the adjusted exposure for the pure premium method will be used to calculate the proposed

→ Approximated (class & average rate determination)

→ Approx. exposure-based credibility over claims count-based when determination of a claim has changed over time

→ Disadvantage of classical credibility is that it requires several simplifying assumptions which may not hold true in some cases

→ Derive base rate ⇒

Variable

Act. exp.

Current rat

Adjusted rat

1

2

3

Summation

total current rat = $\frac{\text{Summation (current rat, adjusted rat)}}{\text{sum (act exp)}}$

total proposed rat = $\frac{\text{Summation (proposed rat, adjusted rat)}}{\text{sum (act exp)}}$

⇒ ODF = $\frac{\text{total current rat}}{\text{total proposed rat}}$

⇒ Proposed base rate = $\frac{\text{proposed rat} - \text{proposed premium}}{\text{current avg premium} - \text{current adjusted rat}} \times \text{current base rate} + \text{ODF}$

→ Extension of exposures (an approximation)

→ Proposed base rate = $\frac{\text{Base rate} \times \frac{\text{Proposed Avg Premium} - \text{Proposed Addition fee}}{\text{Proposed Avg Premium} - \text{Proposed rat base rate}}}{\text{Base rate}}$

can be simplified so

① → Current avg premium = $\frac{\text{rate to exposure}}{\text{exposures}}$

② → Proposed avg premium = (reproposed small rate change) * Current avg premium

③ → Proposed avg premium = same as ②, except replace base rate w/ fixed base rate (frequency: 1)

④ ⇒ Proposed avg base rate = $\frac{\text{Base rate} \times \text{①} - \text{Proposed rat}}{\text{①} - \text{Proposed rat}}$

(CHECK repeat ③ w/ ④ & solve for ① (except total premium)

Base base rate ↑ Proposed base rate

→ Capping rates

→ Method 1: An intuitive, proportional reallocation approach that simplifies adjustments, especially useful for manual-capped portfolios (primarily rework the shortfall)

→ Example →

Class	Proposed Premium	Shortfall Reallocation	Final Premium
A	\$120,778	-\$12,778	\$108,000
B	\$345,081	\$12,778 × \$345,081 / \$517,622 = \$8,519	\$353,600
C	\$172,541	\$12,778 × \$172,541 / \$517,622 = \$4,259	\$176,800
Total	\$638,400	\$0	\$638,400

→ Reallocation of the shortfall in this way effectively alters each of the relativities, not the base rate. We can derive the adjusted relativities by comparing the final premium amounts to the proposed premium amounts.

→ Adjusted relativity = $\frac{\text{Proposed relativity} \times \text{Final premium}}{\text{Proposed premium}}$

→ Benefits → Method 1 is practical & intuitive, avoiding the need for reallocations & providing a streamlined way to achieve the target rate

→ Especially useful in real-world situations w/ multiple capped classes, as it requires only one reallocation & reworking step

→ Drawbacks → May feel less structured than Method 1 due to the allocation process

→ Requires a final reworking step for consistency, taking the base class relativity to 1.00

4.3 -> Introduction to reserves principles

→ Reported claims = Paid claims + Case outstanding

→ Unpaid claims = Case outstanding + IBNR

↳ Brand development

↳ Paid IBNR + IBNR

→ Unpaid claims = $\frac{\text{Paid claims} + \text{Case outstanding} + \text{IBNR}}{\text{Reported claims} + \text{IBNR}}$

4.4 -> Calculations of different reserving methods

→ Development method

→ Based solely on historical data

→ Est ultimate claims = $\frac{\text{Cumulative claims to date} \times \text{CDF}}{\text{Report (or paid)}}$

→ Exposure claims method

→ Determining the ECR (first step)

- 1) Adjust the historical data → This includes reworking the premium, adjusting for legal & environmental changes, developing historical rates, & applying trends (prior to year of interest)
- 2) Calculate the claims ratios → $\text{ECR} = \frac{\text{Ultimate claims (from dev method)}}{\text{reported OLEP}}$
- 3) Calculate average claim ratios (the use different methods)
- 4) Select the ECR → actuarial judgement

→ Use the actual (initial) claim ratio & ignores observed claim experience (i.e. only base calculation on full years of data)

→ Est ultimate claims = $\frac{\text{Expected claims} \times \text{Base premium}}{\text{Expected ECR}}$

↳ OLEP

↳ ECR if no rate change history given

→ Benktander-Pergerson method

→ Deriving blend of the development method & the EC method

→ Split ultimate claims estimate into actual reported (paid) claims & expected reported (unpaid) claims

→ gives more weight to actual claims over time

→ Formula → ① credibility weighted average

Est ultimate claims = $\frac{\text{Development ultimate claims}}{\text{reported}} \times \left(\frac{1}{\text{CDF}} \right) + \frac{\text{Expected claims ultimate claims}}{\text{reported}} \times \left(\frac{1 - \frac{1}{\text{CDF}}}{\text{CDF}} \right)$

* w/ trending here (i.e. just getting est ult claims) & unless have to derive ECR

② Est ultimate claims = $\frac{\text{Actual claims} + \text{Expected claims} \times \text{unreported}}{\text{Report (or paid)}}$

↳ ECR * OLEP

↳ All counts if deriving

↓

accounts for actual YTD report claims

↓

with IBNR view expected claims (i.e. expected a priori loss)

→ Benktander method

→ Essentially an iterative version of the BP technique

→ Benktander ultimate claims = $\frac{\text{Development ultimate claims}}{\text{ultimate claims}} \times \left(\frac{1}{\text{CDF}} \right) + \text{ultimate claims} \times \left(1 - \frac{1}{\text{CDF}} \right)$

→ Cape Cod method

→ ECR is entirely data-driven

→ ① Cape Cod ECR → gives weight to year (not developed) unpaid factor

→ OLEP = $\frac{\text{reported cumulative claims (year or paid)}}{\text{exposure}}$

↳ trend to present ("as of") date

Dev method LDFs

CDF

% reported

used w/ premium

OLEP

w/ report

★ Cape Cod ECR = $\frac{\text{trended cumulative claims}}{\text{used w/ premium}}$

⇒ Expected unreported claims = $\frac{\text{Actual claims}}{\text{ECR}} \times \text{OLEP} \times \text{unreported}$

(or unpaid)

⇒ Est ultimate claims = $\text{Actual claims} + \text{Expected unreported claims}$

↳ Unreported

→ Frequency-severity technique #1

→ Analyze the development technique separately to claim counts & claim severity

→ ① Develop cumulative claim counts ⇒ select ultimate

↳ LDF ⇒ CDF

→ ② Develop cumulative severity ⇒ select ultimate

↳ LDF ⇒ CDF

→ ③ Est ultimate claims = $\frac{\text{ultimate claim count}}{\text{ultimate severity}} \times \text{ultimate severity}$

→ ④ IBNR = $\text{Est ult} - \text{cumulative reported claims}$

↳ = claim count

→ Frequency-severity technique #2

→ Effectively we select a frequency/severity or an average historical ultimate frequency/severity trended to the year we are reserving

→ ① Develop & trend claim count

* trend to an actual date if in-trend

→ ② Trend exposure

→ ③ Calculate ultimate frequencies & select frequency

↳ ①: $\frac{\text{trended ultimate claim count}}{\text{trended exposures}}$

→ ④ Develop & trend severity

⇒ select frequency

→ ⑤ Est ultimate claims = $\frac{\text{AT frequency} \times \text{selected severity}}{\text{selected frequency}} \times \text{selected severity}$

→ ⑥ IBNR = $\text{Est ult} - \text{cumulative reported claims}$

↳ = claim count

→ Frequency-severity technique #3 (Optional rate method)

→ AT Deviation Ratio = $\frac{\text{AT cumulative claim count}}{\text{AT ultimate claim count}}$

→ ① Use incremental paid severity

↳ $\frac{\text{incremental paid claims}}{\text{incremental claim count}}$

→ ② Trend to latest AT & select the claim severity

→ ③ Use deviation ratio & select

(from cumulative claim count)

→ ④ Calculate incremental paid claims, based on reported rates

→ ⑤ Est ultimate claims = $\text{selected severity} \times \text{selected incremental claim count} = \text{Subproduct (above 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 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997, 998, 999, 1000$

⇒ ⑥ ⇒ Est ultimate claims = $\text{actual paid claims} + \text{adjusted paid claims}$

→ Case outstanding technique #1

→ ① Calculate remaining - case ratio

↳ = $\frac{\text{case outstanding}}{\text{case outstanding} + \text{IBNR}}$

→ ② Project case outstanding using